

Software Quality Engineering — Assignment 2

Name: Hira Shahid **Roll Number:** FA23-BSE-029

Topic: Automated Test Suite Development — LMS Fine Calculator

This document contains the specification-based test design, traceability matrix, defect evidence, and reflection required by Tasks 1 and 5.

Techniques: Boundary Value Analysis, Equivalence Partitioning, and Decision Table testing.

Task 1 — Test Design Table

TC ID	Spec Rule	Technique	Input	Expected Output
TC-01	R3,R4,R5,R6	Equivalence partitioning	(10, "student", "book")	4.00
TC-02	R3,R4,R5,R6	Equivalence partitioning	(5, "guest", "media")	6.00
TC-03	R3,R4,R5,R6	Equivalence partitioning	(6, "staff", "reserve")	4.00
TC-04	R3	Boundary value analysis	(-1, "student", "book")	0.00
TC-05	R3	Boundary value analysis	(0, "student", "book")	0.00
TC-06	R3	Boundary value analysis	(2, "student", "book")	0.00
TC-07	R3	Boundary value analysis	(3, "student", "book")	0.50
TC-08	R4,R5	Decision table	(10, "student", "book")	4.00
TC-09	R4,R5	Decision table	(10, "staff", "book")	2.00
TC-10	R4,R5	Decision table	(10, "guest", "book")	8.00
TC-11	R4,R5	Decision table	(10, "student", "media")	8.00
TC-12	R4,R5	Decision table	(10, "staff", "media")	4.00
TC-13	R4,R5	Decision table	(10, "guest", "media")	16.00
TC-14	R4,R5	Decision table	(10, "student", "reserve")	16.00
TC-15	R4,R5	Decision table	(10, "staff", "reserve")	8.00
TC-16	R4,R5	Decision table	(10, "guest", "reserve")	32.00
TC-17	R7	Boundary value analysis	(110, "student", "book")	50.00
TC-18	R7	Boundary value analysis	(110, "staff", "book")	27.00
TC-19	R7	Boundary value analysis	(110, "guest", "book")	50.00
TC-20	R7	Boundary value analysis	(110, "student", "media")	50.00
TC-21	R7	Boundary value analysis	(110, "guest", "media")	50.00
TC-22	R7	Boundary/decision table	(110, "student", "reserve")	216.00
TC-23	R7	Boundary/decision table	(110, "staff", "reserve")	108.00
TC-24	R7	Boundary/decision table	(110, "guest", "reserve")	432.00
TC-25	R2	Equivalence partitioning	("5", "student", "book")	TypeError
TC-26	R2	Equivalence partitioning	(2.5, "student", "book")	TypeError
TC-27	R2	Equivalence partitioning	(None, "student", "book")	TypeError
TC-28	R1	Equivalence partitioning	(5, "alumni", "book")	ValueError
TC-29	R1	Equivalence partitioning	(5, "admin", "book")	ValueError
TC-30	R1	Equivalence partitioning	(5, "", "book")	ValueError
TC-31	R1	Equivalence partitioning	(5, None, "book")	ValueError
TC-32	R1	Equivalence partitioning	(5, "student", "dvd")	ValueError
TC-33	R1	Equivalence partitioning	(5, "student", "magazine")	ValueError
TC-34	R1	Equivalence partitioning	(5, "student", "")	ValueError
TC-35	R1	Equivalence partitioning	(5, "student", None)	ValueError

TC ID	Spec Rule	Technique	Input	Expected Output
TC-36	R8	Boundary value analysis	fine=0.00	{"sent": False, "error": "Fine below notice threshold"}
TC-37	R8	Boundary value analysis	fine=4.99	{"sent": False, "error": "Fine below notice threshold"}
TC-38	R8	Boundary value analysis	fine=5.00	{"sent": False, "error": "Fine below notice threshold"}
TC-39	R8,R10	Boundary value analysis	fine=5.01	{"sent": True, "error": None}
TC-40	R8,R10	Boundary value analysis	fine=6.00	{"sent": True, "error": None}
TC-41	R8,R10	Boundary value analysis	fine=50.00	{"sent": True, "error": None}
TC-42	R9	Exception + mock testing	fine=6.00; _deliver_email raises ConnectionError("server unavailable")	{"sent": False, "error": "SMTP error: server unavailable"}
TC-43	R10	Mock-based integration check	_deliver_email(student@example.com, 6.25)	SMTP message created and send_message called

Task 1b — Traceability Matrix

Rule	Covered by TC IDs	Coverage meaning
R1	TC-28–TC-35	Valid/invalid member and item partitions
R2	TC-25–TC-27	Non-integer days partitions
R3	TC-01–TC-07	Normal and grace-period boundary cases
R4	TC-01–TC-03, TC-08–TC-16	All item daily-rate categories
R5	TC-01–TC-03, TC-08–TC-16	All member multipliers
R6	TC-01–TC-03, TC-08–TC-16	Calculated fine examples
R7	TC-17–TC-24	Cap and no-cap reserve cases
R8	TC-36–TC-41	Threshold below/equal/above 5.00
R9	TC-42	ConnectionError handling
R10	TC-39–TC-41, TC-43	Successful notice and SMTP delivery

Task 3a: 17 normal, 15 boundary, and 11 invalid cases, giving 43 total pytest cases. Parametrized rows are counted individually.

Task 4 — Defect Detection

Failing Test	Expected	Actual	Defect Found
TC-18 — staff/book at 110 days	27.00	25.00	R7: cap was applied before the staff multiplier.
TC-19 — guest/book at 110 days	50.00	100.00	R7: cap was applied before the guest multiplier.
TC-21 — guest/media at 110 days	50.00	100.00	R7: cap was applied before the guest multiplier.
TC-38 — notice at fine 5.00	sent=False; threshold message	sent=True	R8: implementation used < 5.00 instead of requiring fine > 5.00.

Corrected result: 43 passed. Coverage.py reports 100% statement coverage and 100% branch coverage. The --fail-under=90 command returns exit code 0.

Task 5 — Reflection

The test suite uses three main techniques from the assignment. Equivalence partitioning produced tests for valid and invalid member types, item types, and days_overdue types. Boundary Value Analysis was used around the two-day grace period, the 50.00 cap, and the 5.00 notice threshold. The decision-table tests cover the combinations of the three member types and three item types. The coverage result is 100% for both statements and branches, but coverage percentage does not prove that the requirements are correct or that every output is meaningful. A line can execute without an assertion checking the exact business result. This is why the suite uses specific assertions for fine amounts, return dictionaries, and mocked email calls instead of relying only on execution coverage. The suite is sufficient for the specified rules because every R1–R10 rule is traced to at least one test, normal, boundary, and invalid cases are included, and the original implementation's two defects are exposed. With more time, I would add randomized/property-based tests, more rounding combinations, and additional email-validation cases if those behaviors were added to the specification.

Evidence files in the ZIP include the original-code failures, corrected-code passing run, and coverage report.